Shape of $SnCl_2$

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level P1

Which of the following structures best represents the gaseous $SnCl_2$ molecule? The orbital lobe represents a lone (unshared) pair of electrons.

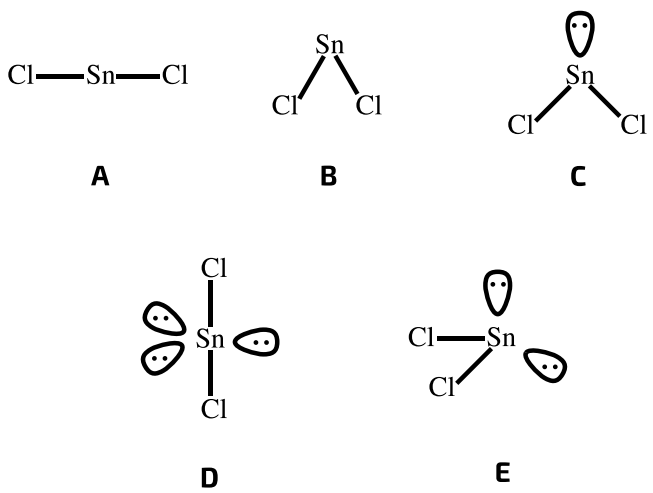


Figure 1: Possible shapes of $SnCl_2$

- A
- B
- C
- D
- E

Adapted with permission from UCLES, A-Level Chemistry, November 1991, Paper 1, Question 5

Question deck:
STEM SMART Chemistry Week 7

Shapes and Angles

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level P1

Part A
BCl_3 and PCl_3

Why (according to VSEPR theory) is the molecule of BCl_3 planar, whereas the molecule of PH_3 is pyramidal?

- The boron atom has no d-orbitals available for bonding.
- The boron atom in BCl_3 has six electrons in its valency shell, whereas the phosphorus atom in PH_3 has eight.
- The repulsion between chlorine atoms is greater than that between hydrogen atoms.
- The covalent radius of phosphorus is greater than that of boron.
- The covalent radius of chlorine is greater than that of hydrogen.

Part B
NH_3

In the ammonia molecule, what is the approximate value of the $H-N-H$ bond angle?

- $180°$
- $120°$
- $107°$
- $90°$
- $60°$

Part A adapted with permission from UCLES, A-Level Chemistry, June 1991, Paper 3, Question 4;
Part B adapted with permission from OCSEB, A-Level Chemistry, June 1994, Paper 1, Question 1

Question deck:
STEM SMART Chemistry Week 7

Bond Angles

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level P1

Part A

Methane, ammonia and water

The bond lengths and bond angles in the molecules of methane, ammonia and water may be represented as follows:

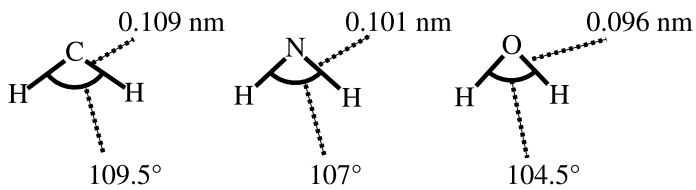


Figure 1: Shapes of molecules of methane, ammonia and water.

What causes this trend in the bond angles shown, according to valence shell electron pair repulsion theory?

1 increasing repulsion between hydrogen atoms as the bond length decreases
2 the number of non-bonding electron pairs in the molecule
3 a nonbonding electron pair having a greater repulsive force than a bonding electron pair

- **1, 2** and **3** are correct
- **1** and **2** only are correct
- **2** and **3** only are correct
- **1** only is correct
- **3** only is correct

Part B

SO_3^{2-}

The SO_3^{2-} ion may be represented as (geometry not necessarily representative):

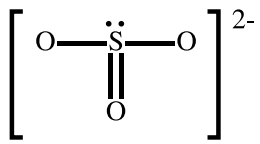


Figure 2: SO_3^{2-} ion

What is the $O-S-O$ bond angle?

- 90° exactly
- about 107°
- about 109.5°
- 120° exactly

Part A adapted with permission from UCLES, A-Level Chemistry, June 1992, Paper 4, Question 31;
Part B adapted with permission from UCLES, A-Level Chemistry, November 1993, Paper 4, Question 2

Question deck:

STEM SMART Chemistry Week 7

Dative Bond and Similar Shapes

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level P1

Part A

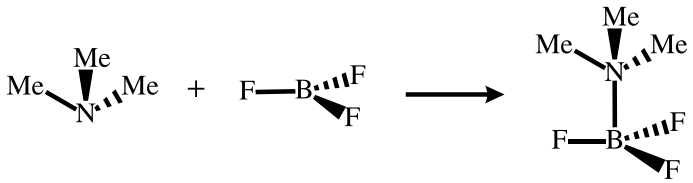
Me_3N and BF_3

Trimethylamine, Me_3N, reacts with boron trifluoride, BF_3, to form a compound of formula Me_3NBF_3.

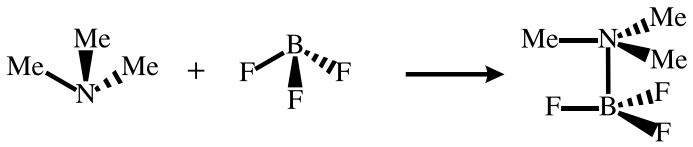
$[Me = CH_3]$

How may this reaction be drawn in terms of the shapes of the reactants and products?

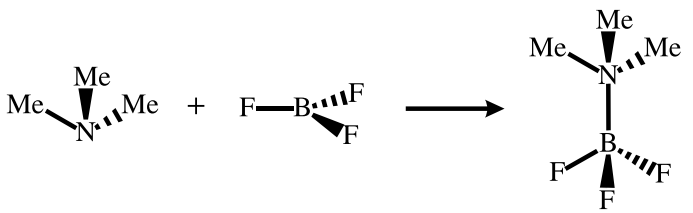
A



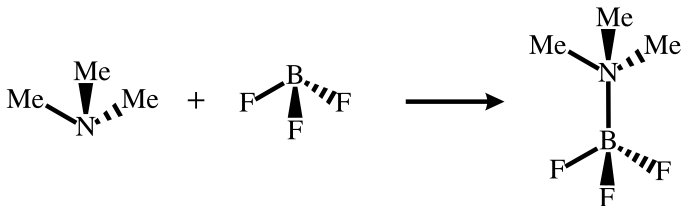
B



C



D



- A
- B
- C
- D

Part B

Similar shapes

In which of the following pairs do the molecules have similar shapes?

- $AlCl_3$ and BCl_3
- $AlCl_3$ and PCl_3
- BF_3 and NH_3
- $BeCl_2$ and H_2O
- CO_2 and SO_2

Part A adapted with permission from UCLES, A-Level Chemistry, June 1995, Paper 4, Question 3;
Part B adapted with permission from UCLES, A-Level Chemistry, June 1993, Paper 4, Question 6

Question deck:
STEM SMART Chemistry Week 7

Bonding and Shapes

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level P1

Part A

$BF_3 \cdot CH_3OH$

$BF_3 \cdot CH_3OH$ is a reagent used to form methyl esters from compounds containing acyl groups. In the diagrams, x, • and ○ represent electrons from B, F and O, respectively.

Which tetrahedral structure illustrates the electron pairs around the boron atom?

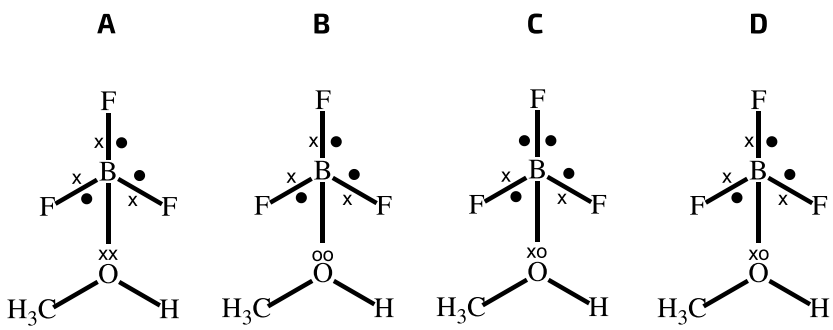


Figure 1: Possible distributions of electrons in $BF_3 \cdot CH_3OH$

- A
- B
- C
- D

Part B

Not planar

Which of the following molecules is not planar?

- Phosphorus trichloride
- Ethene
- Xenon tetrafluoride
- Boron trifluoride

Part A adapted with permission from UCLES, A-Level Chemistry, November 1997, Paper 3, Question 2;
Part B adapted with permission from UCLES, A-Level Chemistry, November 1997, Paper 3, Question 6

Question deck:
STEM SMART Chemistry Week 7

Shapes of Fluorides

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level P1

For each of the following, enter a one to two word answer, using appropriate shape of molecule terminology, e.g. "linear".

Part A

BF_3

Describe the shape of BF_3.

Part B

CF_4

Describe the shape of CF_4.

Part C

NF_3

Describe the shape of NF_3.

Part D

SF_6

Describe the shape of SF_6.

Part A adapted with permission from UCLES, A-Level Chemistry, November 1995 , Paper 1, Question 1

Question deck:

STEM SMART Chemistry Week 7

Shapes of Halide Compounds

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level C1

For each of the following, deduce the shape of the molecules and enter a one to two word answer, using appropriate shape of molecule terminology, e.g. "linear".

Part A

BBr_3

Deduce the shape of BBr_3.

Part B

PF_3

Deduce the shape of PF_3.

Part C

SF_4

Deduce the shape of SF_4.

Part D

IF_5

Deduce the shape of IF_5.

Part E

$AlCl_3$ and Cl^-

Predict the shape of the species formed from the reaction of $AlCl_3$ with Cl^-.

Adapted with permission from OCSEB, STEP Chemistry, Jun 1998, Question 4

Question deck:

STEM SMART Chemistry Week 7

Shapes of Xenon Compounds

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level C2

For each of the following, deduce the shape of the molecules and enter a one to two word answer, using appropriate shape of molecule terminology, e.g. "linear".

Part A

XeF_2

Describe the shape of XeF_2.

Part B

$XeOF_2$

Describe the shape of $XeOF_2$.

Part C

XeO_4

Describe the shape of XeO_4.

Part D

XeF_4

Describe the shape of XeF_4.

Part E

$XeOF_4$

Describe the shape of $XeOF_4$.

Part A adapted with permission from OCR, STEP Chemistry, June 1999, Question 5

Question deck:

STEM SMART Chemistry Week 7

Shapes of Molecules and Ions

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level P1

Part A

F_2O

By considering the number of lone and bonding pairs of electrons, predict the shape of F_2O.

Part B

H_3O^+

By considering the number of lone and bonding pairs of electrons, predict the shape of H_3O^+.

Part C

ClF_4^-

By considering the number of lone and bonding pairs of electrons, predict the shape of ClF_4^-.

Part D

$SbF_5{}^{n-}$

Antimony, Sb, is in group 15 of the Periodic Table. It forms a series of salts which contain the $SbF_5{}^{n-}$ anion, the structure of which is a square-based pyramid:

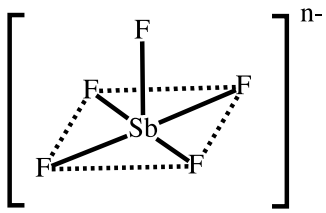


Figure 1: Structure of the $SbF_5{}^{n-}$ anion

Deduce the total number of electrons around the antimony atom.

Deduce the value of n.

Adapted with permission from UCLES, A-Level Chemistry, June 1991, Paper 3, Question 2

Question deck:

STEM SMART Chemistry Week 7

Shapes of Molecules and Ions Extension

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level C1

Antimony, Sb, is in group 15 of the Periodic Table. It forms a series of salts which contain the $\mathrm{SbF_5}^{n-}$ anion, the structure of which is a square-based pyramid:

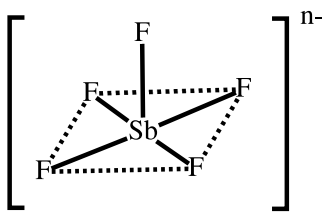


Figure 1: Structure of the $\mathrm{SbF_5}^{n-}$ anion

Deduce the oxidation number of Sb in the $\mathrm{SbF_5}^{n-}$ anion above.

Adapted with permission from UCLES A-Level Chemistry June 1991, Paper 3, Q2

Question deck:
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